

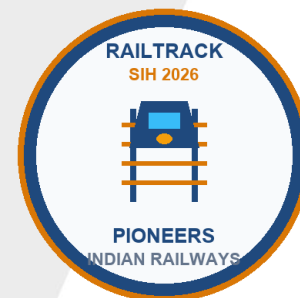
# SMART INDIA HACKATHON 2026

## TITLE PAGE



## PAGE

- Problem Statement ID – **SIH26028**
- Problem Statement Title- **AI-Driven Dynamic Section Controller & Track Maintenance Block Scheduling System**
- PS Category- **Software**
- Theme- Transportation & Logistics / Smart Railways
- Team ID- SIH-26028-TEAM
- Team Name (Registered on portal) : **RailTrack Pioneers**





# AI-Driven Section Controller & Maintenance Block Planner

## ❖ Proposed Solution



### We Propose:

- **Autonomous Block Scheduling:** Constraint Programming (Google OR-Tools CP-SAT) dynamically computes conflict-free 2-4 hour track possession windows without cancelling express trains.
- **Kinematic Conflict Gating:** Continuous GPS/NavIC tracking with mathematical 5-minute safety headway enforcement between trains and heavy track machines.
- **Multi-Model ETA Predictor:** SLSQP + XGBoost meta-ensemble predicting dynamic section arrival times with sub-2-minute accuracy.

### Uniqueness:

- **Closed-Loop Dual Control:** Section Controller Gantt workbench + passenger transparency portal.
- **Dynamic IMD Weather Guardrail:** Automatically expands safety buffers (+5m/+10m) upon adverse rain/fog alerts.

### • Mathematical G&SR Compliance:

Enforces Indian Railways Block Working Rules and automatic signaling blocks.

### • Sub-1.2s Optimization Speed:

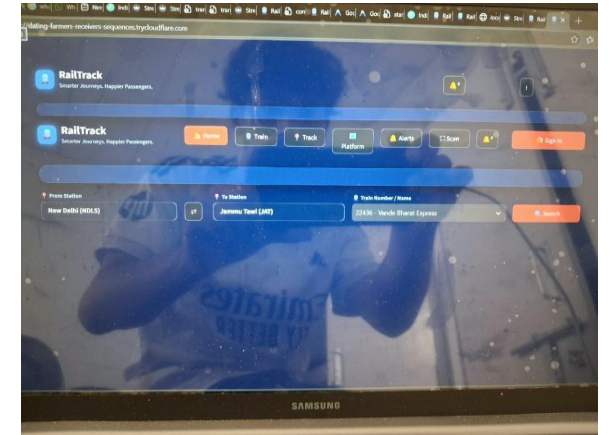
Evaluates 40+ track parameters simultaneously on standard laptops.

### • Local SQLite WAL Continuity:

Ensures 100% offline continuity during fiber or WAN connectivity failures.

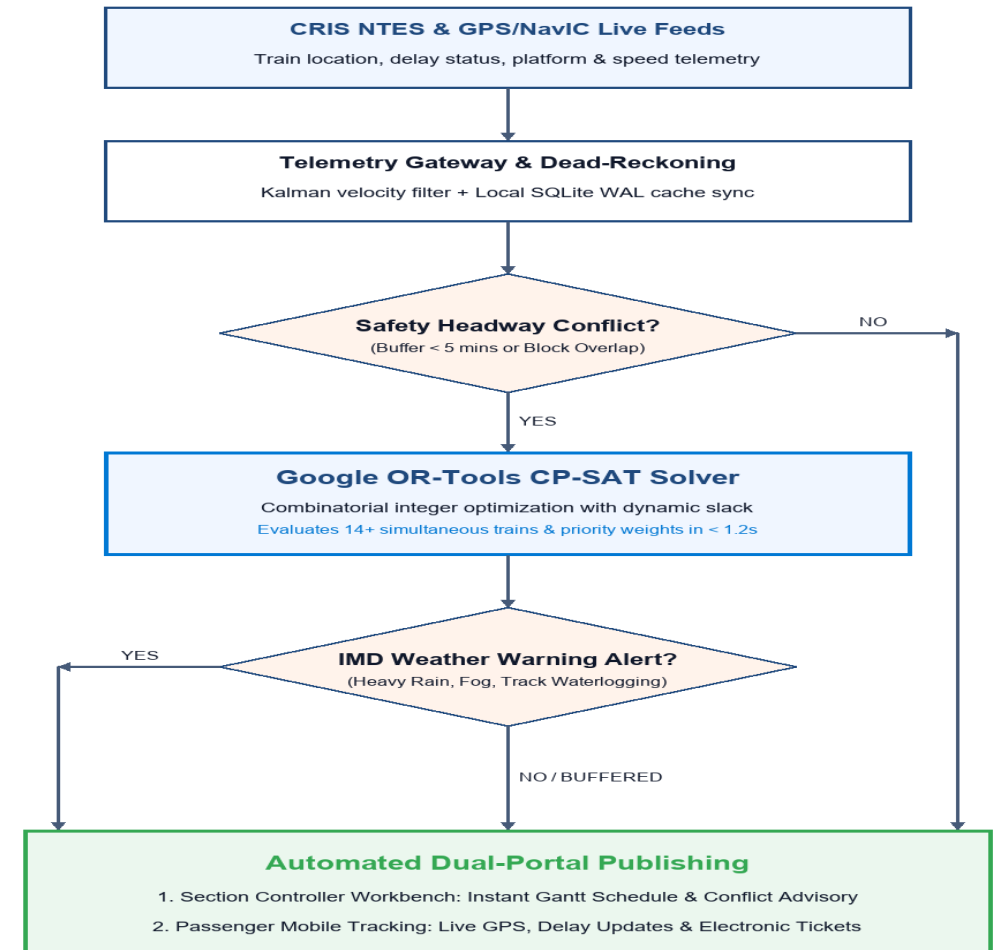
### • Multi-Corridor Scalability:

Tested on high-density New Delhi – Ludhiana – Jammu Tawi section.



## Core Technology Stack & Architecture

- **Optimization Engine:** Google OR-Tools 9.10 (CP-SAT Solver) with integer linear constraints and dynamic slack.
- **Backend Services:** FastAPI (Python 3.11), Uvicorn ASGI, REST APIs, WebSockets for live telemetry broadcast.
- **Machine Learning:** Dynamic ETA Predictor: Meta-Ensemble (SLSQP, XGBoost, Random Forest) trained on IR timetable data.
- **Data Ingestion:** CRIS NTES connector, Open-Meteo REST API, and IMD Weather Telemetry client with fallback cache.
- **Database & Persistence:** Supabase Cloud PostgreSQL with resilient local SQLite WAL offline auto-sync.
- **Frontend & Maps:** Streamlit responsive mobile/desktop dashboard, Plotly interactive OpenStreetMap GIS tracking.





# FEASIBILITY AND VIABILITY



## Challenges & Solutions

**Operational Feasibility:** Controller workflow requires zero retraining.

Fits standard Control Office Application (COA) layout.

Replaces chaotic phone calls with single-click advisory.

Supports dynamic maintenance window adjustments.

**Technical Feasibility:** FastAPI + Google OR-Tools + Scikit-Learn.

Sub-1.2s optimization solves full section schedules.

100% offline continuity with SQLite WAL database.

Seamless integration with ISRO NavIC GPS telemetry.

**Economic Feasibility:** Zero trackside hardware procurement cost.

Uses existing locomotive GPS & CRIS NTES feeds.

Recovers ₹100s of crores in avoided idle fuel.

Maximizes return on high-speed rail corridors.

- GPS drift & telemetry packet loss in remote railway terrains

- Sudden extreme monsoon rains and severe winter fog

- Congested mixed traffic: high-speed Vande Bharat vs freight

- High track worker safety risks during emergency possessions

- Kinematic dead-reckoning filter with local SQLite WAL sync

- Automated IMD weather guardrail injecting dynamic buffers (+5m/+10m)

- Priority-weighted CP-SAT solver protects high-priority express slots

- Cryptographic token commit gating before track section lock

## Potential Impact

- 85% Reduction in track possession planning overhead.
- Prevents unplanned train halts & freight siding delays.
- Transparent passenger live tracking & verified PNR status.

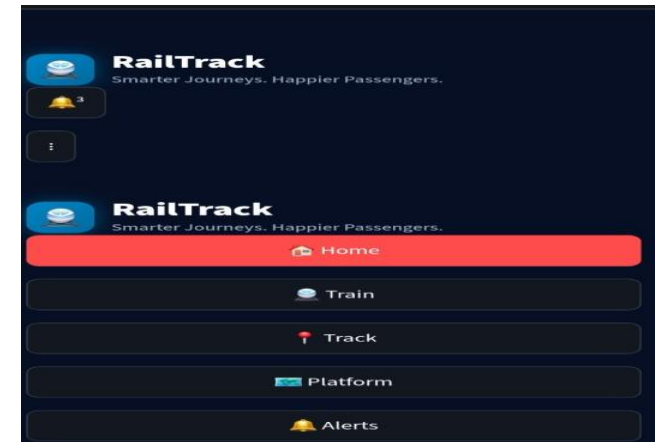
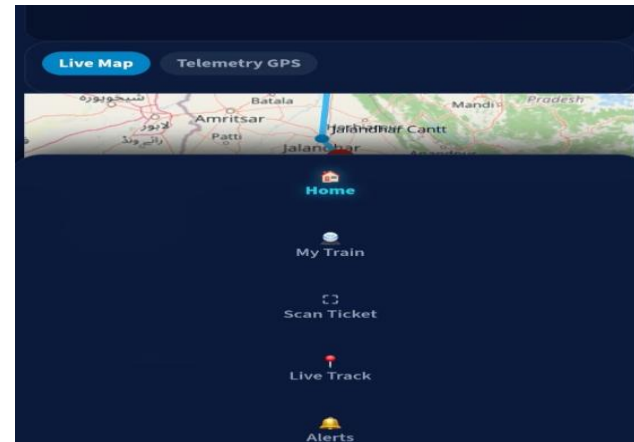
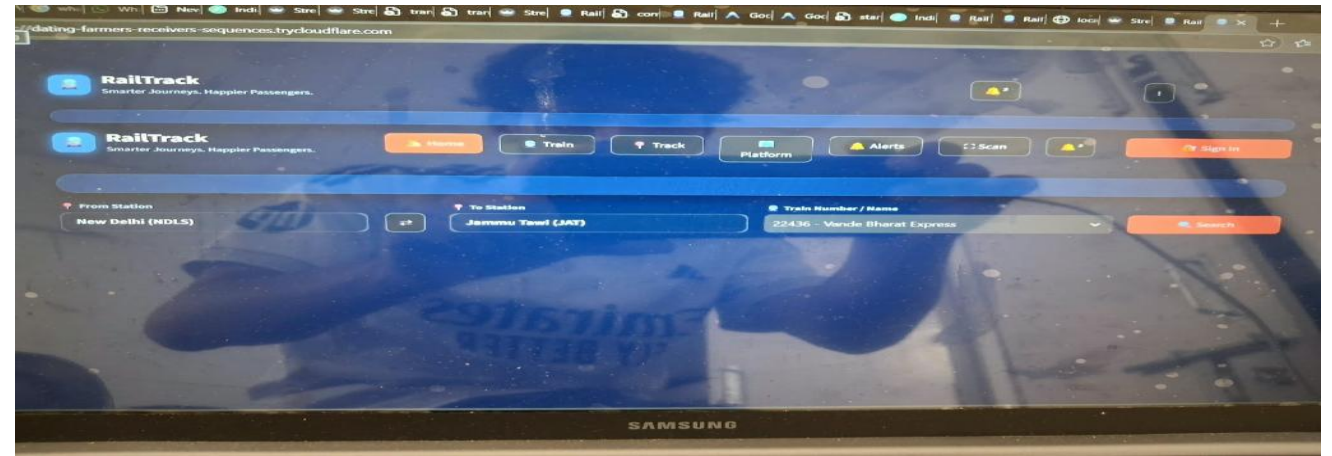
## Benefits:

**Economic:** Boosts line capacity from 70% to 92%, saving ₹100s of crores in diesel idling & train regulation costs.

**Safety & Social:** Guarantees mathematical 5-min safety headway; completely prevents maintenance collisions.

**Environmental:** Substantially cuts standing diesel locomotive idling and brake wear, lowering carbon footprint.

## PROTOTYPE IMAGES (LIVE RAILTRACK APPLICATION)





# RESEARCH AND REFERENCES



- **Ministry of Railways, Government of India (2024).** Indian Railways General & Subsidiary Rules (G&SR) and Block Working Manual for Automatic Block Signaling Corridors. Railway Board.
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- **Chen, T., & Guestrin, C. (2016).** XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.